

## A Decentralized Federated Learning Framework via Committee Mechanism with Convergence Guarantee

Chunjiang Che<sup>1</sup>, Xiaoli Li<sup>1</sup>, Chuan Chen<sup>1</sup>  
Xiaoyu He<sup>1</sup>, and Zibin Zheng<sup>2</sup>

**Speaker: Chunjiang Che**

1. School of Computer Science, Sun Yat-sen University
2. School of Software Engineering, Sun Yat-sen University



中山大學  
SUN YAT-SEN UNIVERSITY

# Outline

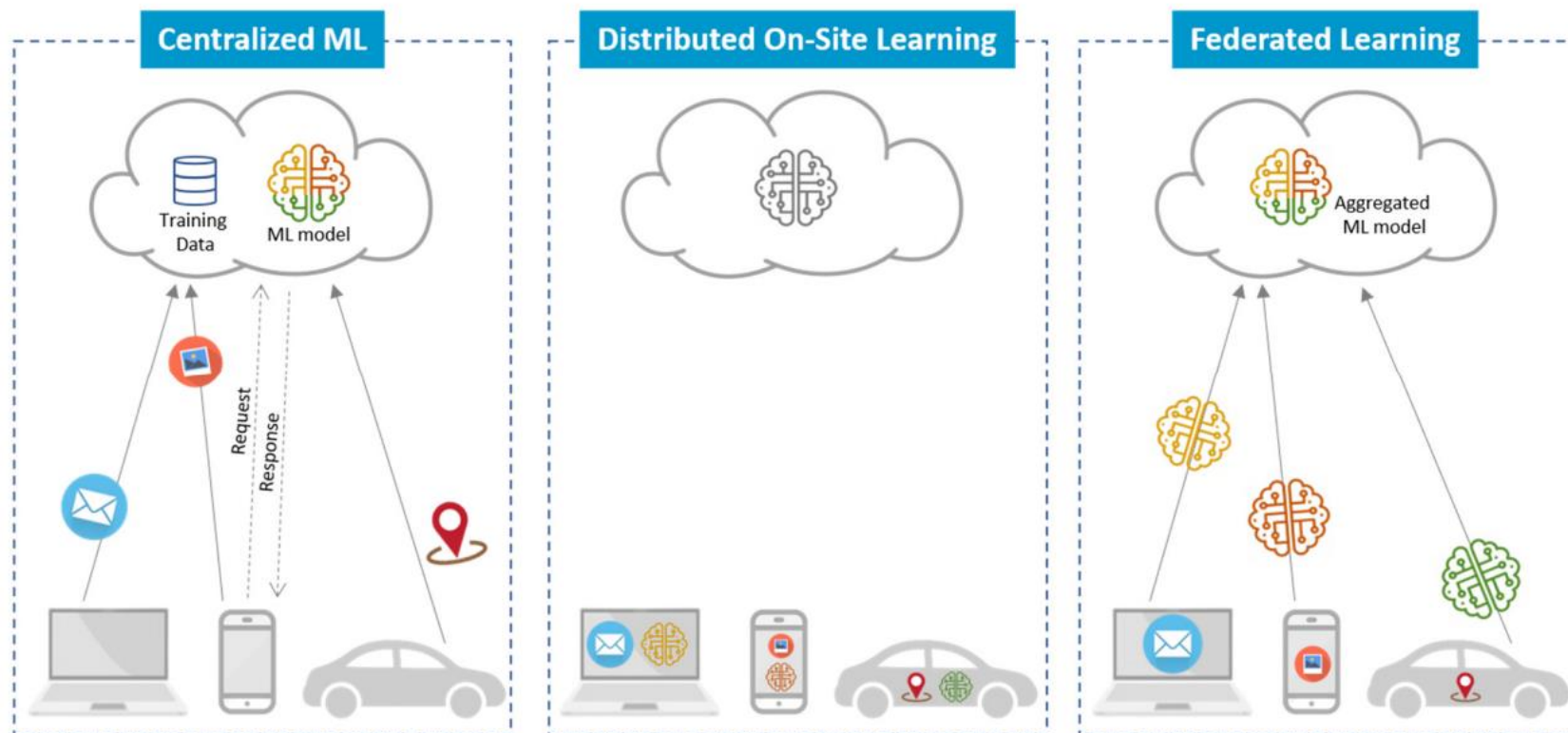


- Background
- Problem Formulation
- Framework
- Theoretical Analysis
- Experimental Results
- Proof of Theorem
- Conclusion

# Distributed Machine Learning



- Centralized ML method violates user privacy
  - Federated Learning protects user privacy

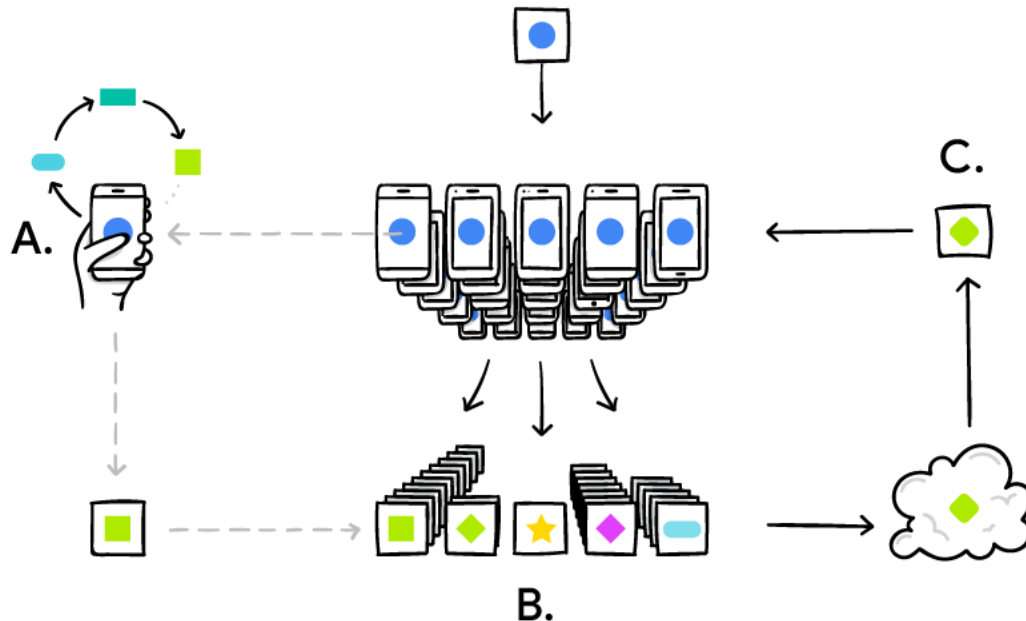


# Federated Learning



## ■ Algorithm

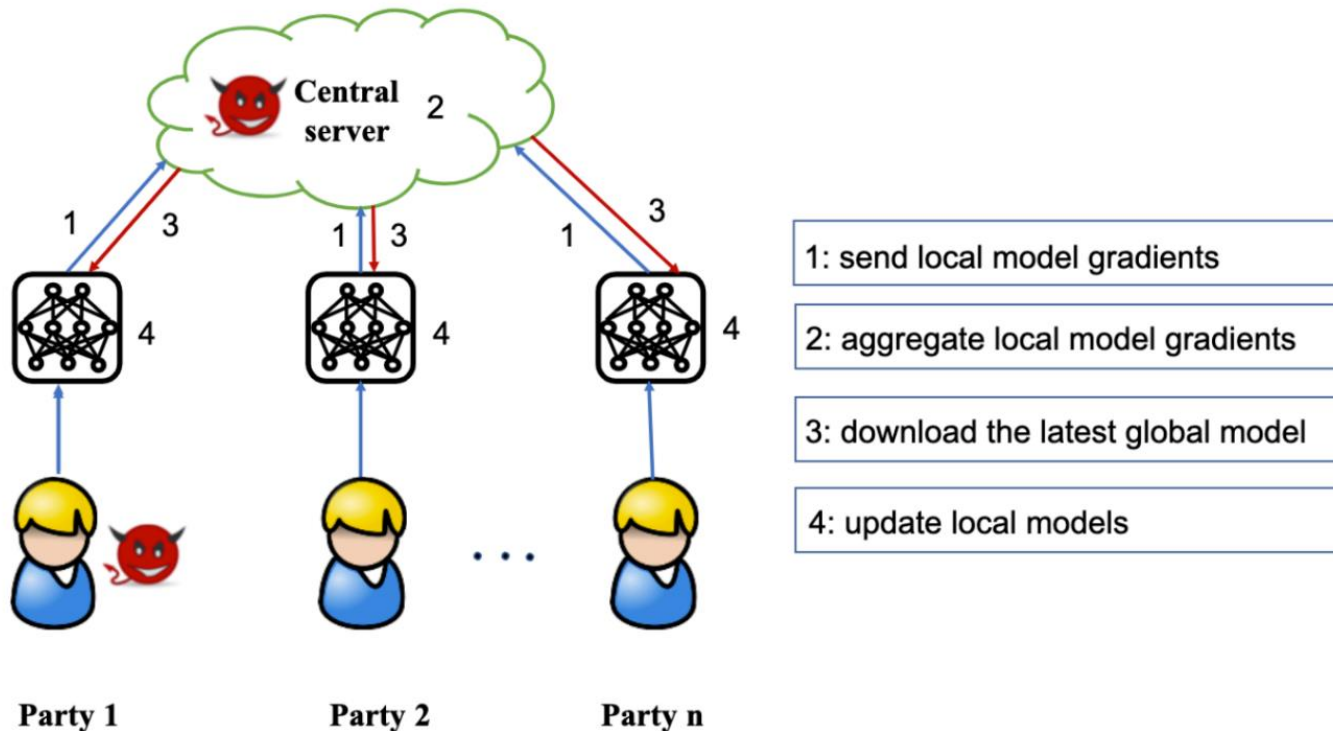
- Local training, gradient uploading, aggregation, model downloading



# Malicious Participant



- Byzantine attacks reduce model performance
  - Current Byzantine-tolerance algorithms: Median, Trimmed Mean, Krum and Multi-Krum

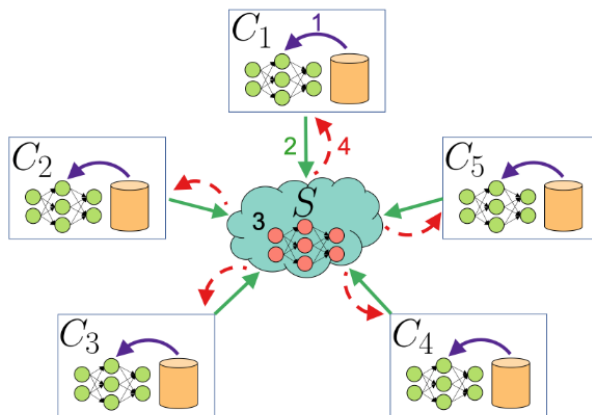


# Decentralized Federated Learning

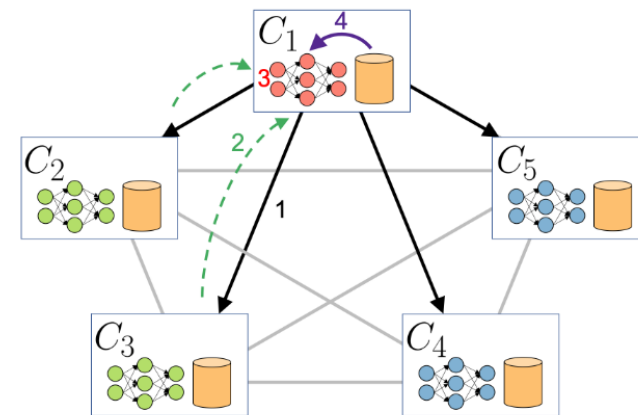


## ■ BrainTorrent

- Nodes complete aggregation through P2P protocol



(a) Federated Learning with server



(b) BrainTorrent: P2P serverless Federated Learning

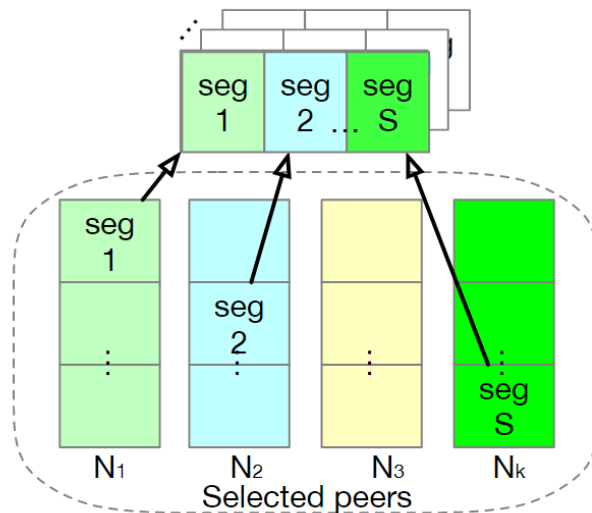


# Decentralized Federated Learning

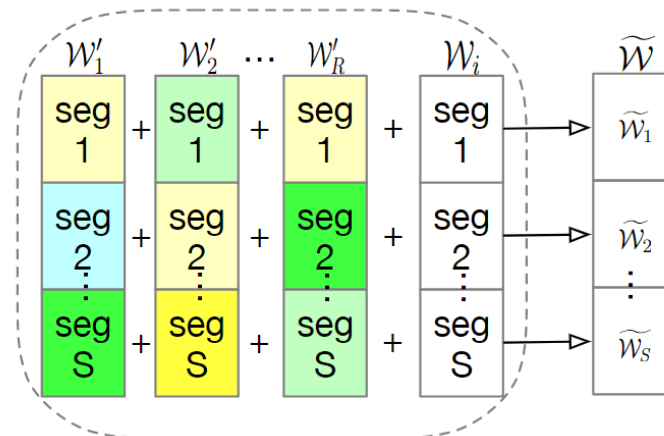


## ■ Gossip

- Nodes complete aggregation through Gossip protocol



(a) Segmented Pulling



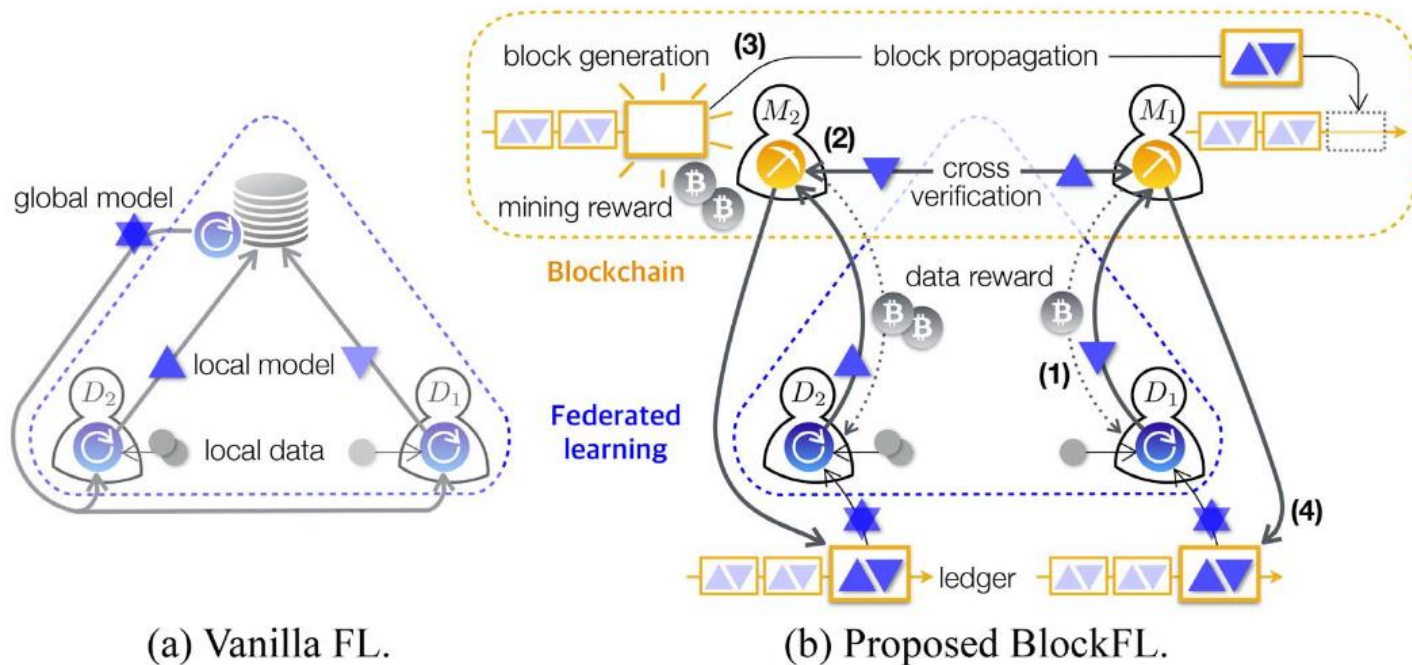
(b) Segmented Aggregation

# Decentralized Federated Learning



## ■ Blockchain

- Nodes complete aggregation through Blockchain consensus



# Limitation



- Current Byzantine-tolerance algorithms
  - Cannot achieve satisfactory performance
  - Cannot provide effective defense in the presence of malicious servers
- Current decentralized FL frameworks
  - Not resistant to Byzantine attacks
  - High communication overhead

CMFL: A Robust Decentralized  
Federated Learning Framework

# Outline



- Background
- Problem Formulation
- Framework
- Theoretical Analysis
- Experimental Results
- Proof of Theorem
- Conclusion

# Problem Formulation



- Local objective function

$$F_k(\mathbf{w}) = \frac{1}{n_k} \sum_{\xi \in \mathcal{D}_k} f(\mathbf{w}, \xi).$$

- Global objective function

$$F(\mathbf{w}) = \sum_{k=1}^K p_k F_k(\mathbf{w})$$

- Local SGD training

$$g_k(\mathbf{w}_{k,i}^t, \mathcal{B}_{k,i}^t) = \frac{1}{|\mathcal{B}_{k,i}^t|} \sum_{\xi \in \mathcal{B}_{k,i}^t} \nabla f(\mathbf{w}_{k,i}^t, \xi).$$

# Problem Formulation



- Local update

$$\mathbf{w}_{k,i+1}^t = \mathbf{w}_{k,i}^t - \eta_i^t g_k(\mathbf{w}_{k,i}^t, \mathcal{B}_{k,i}^t)$$

- Aggregation

$$\bar{g}^t = \sum_{k \in S^t} p_{k,S^t} g_k(\mathbf{w}_{k,\tau}^t, \mathcal{B}_{k,\tau}^t).$$

- Global update

$$\bar{\mathbf{w}}^{t+1} = \bar{\mathbf{w}}^t - \eta_t \bar{g}^t.$$

# Outline

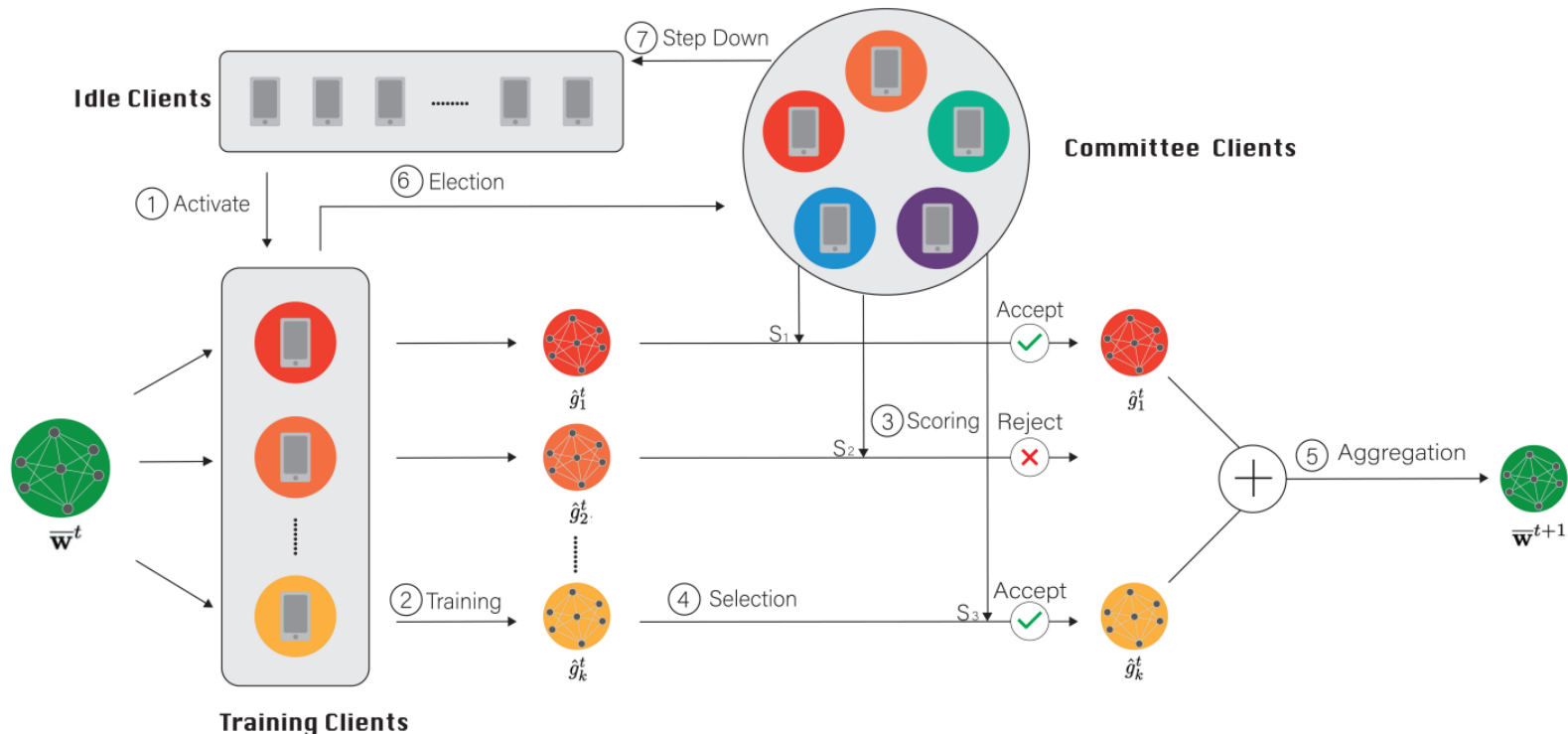


- Background
- Problem Formulation
- Framework
- Theoretical Analysis
- Experimental Results
- Proof of Theorem
- Conclusion

# Committee Mechanism based FL



- CMFL utilize the committee mechanism to implement a decentralized FL framework



# Scoring System



- The scoring system based on Euclidean distance to distinguish malicious and honest gradients
  - Honest gradients get a **high** score
  - Malicious gradients get a **low** score

$$\mathcal{P}_k^c = \frac{1}{\|\hat{g}_k^t - \hat{g}_c^t\|_2^2}.$$



The c-th committee  
client's score

$$\mathcal{P}_k = \frac{1}{\frac{1}{C} \sum_c^C \|\hat{g}_k^t - \hat{g}_c^t\|_2^2} = \frac{C}{\sum_c^C \frac{1}{\mathcal{P}_k^c}}$$



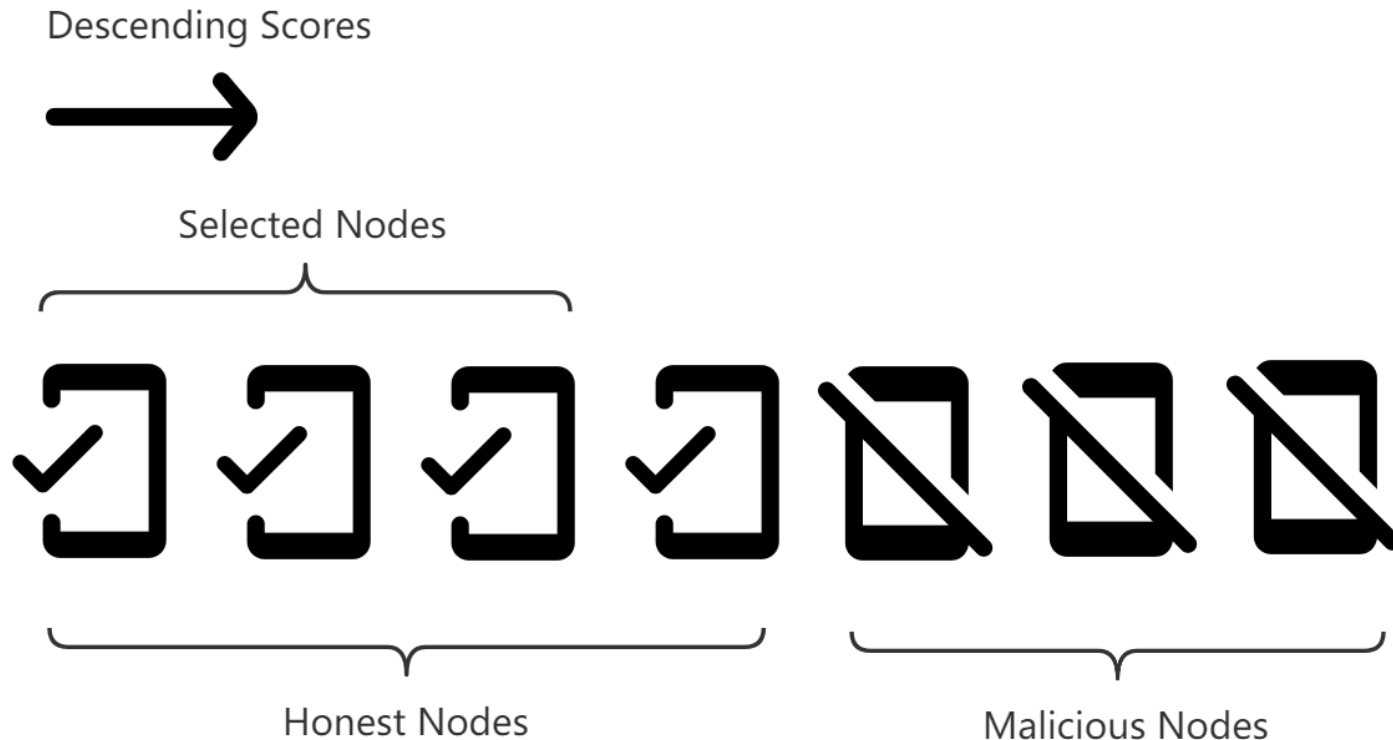
Total score

# Selection Strategy



## ■ Selection strategy I

- Consideration: Robustness
- Applicable scenario: Byzantine attack scenario



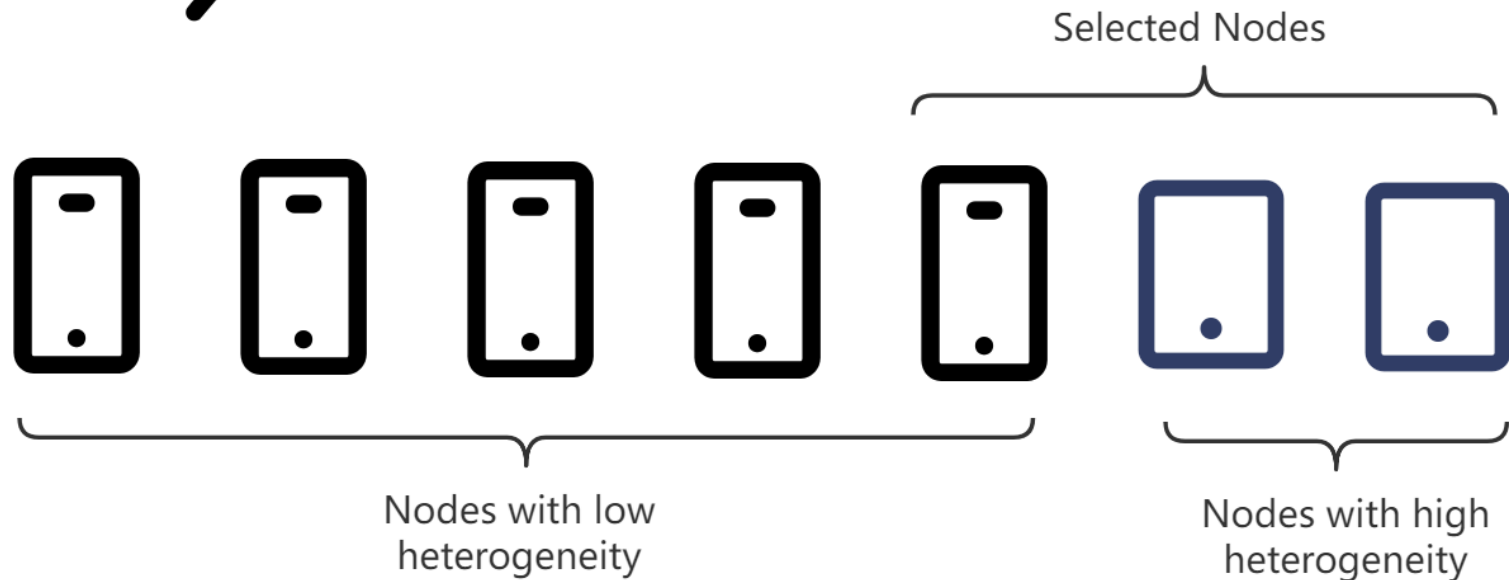
# Selection Strategy



## ■ Selection strategy II

- Consideration: Heterogeneity
- Applicable scenario: Non-attack scenario

Descending Scores



# Election Strategy



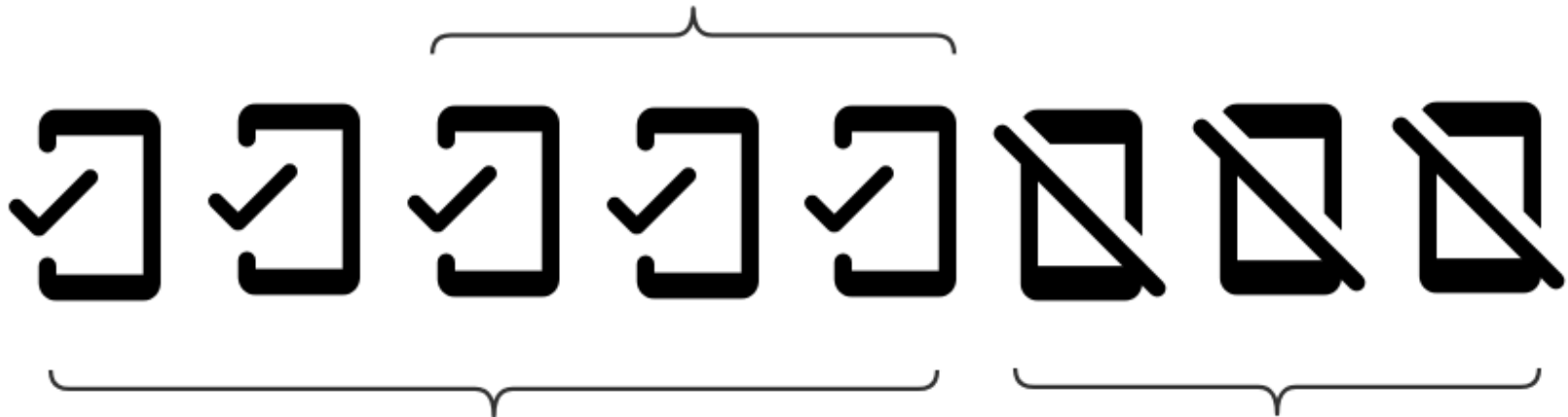
## ■ Considerations

- Robustness
- System Stability

Descending Scores



Selected Nodes



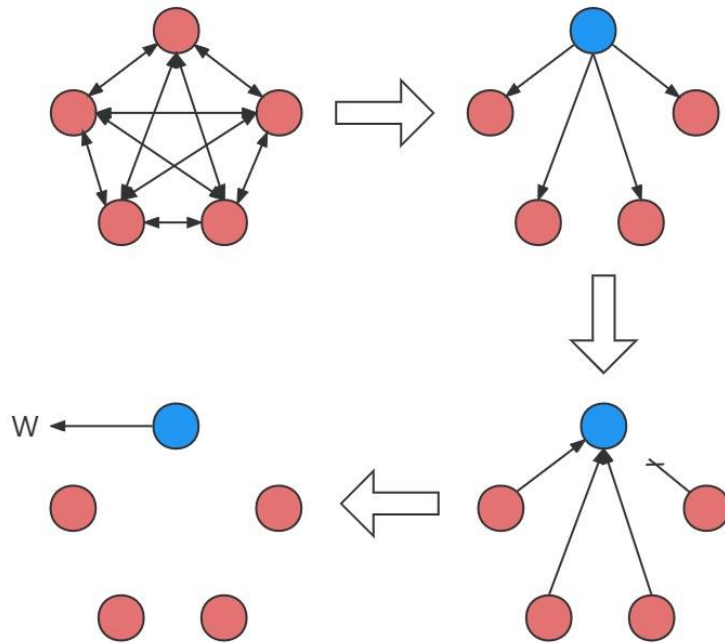
Honest Nodes

Malicious Nodes

# Committee Consensus Protocol



- The committee members reach a consensus by **voting**.



## Algorithm 1 Committee Consensus Protocol

**Input:**  $t, S_c^t, S_b^t$   
**Output:** global model  $w^t$ , new committee client set  $S_c^{t+1}$

```

1: for  $c \in S_c^t$  do
2:   for  $k \in S_b^t$  do
3:     The  $c$ -th committee client broadcasts score  $\mathcal{P}_k^c$  to other committee clients.
4:   end for
5: end for
6: for  $c \in S_c^t$  do
7:   for  $k \in S_b^t$  do
8:     The  $c$ -th committee client calculates the total score  $\mathcal{P}_k$  of the  $k$ -th training client.
9:   end for
10: end for
11: The committee performs voting( $S_a^t$ , aggregation) as Algorithm 2.
12: The committee performs voting( $S_c^{t+1}$ , None) as Algorithm 2.

```

## Algorithm 2 The voting algorithm

**Input:** The client set  $S$  and the operation  $o$

```

1: while No consensus on  $S$  do
2:   The primary committee client  $p$  is selected from committee clients by random.
3:   for  $k \in S_c^t$  do
4:     The  $k$ -th committee client decides its client set  $S^k$  according to the corresponding strategy (selection strategy/election strategy).
5:   end for
6:    $p$  performs the operation  $o$ .
7:    $p$  creates a request  $\langle \text{Request}, S^p, \text{operation}, \text{timestamp} \rangle$  and broadcasts it to all the replicate committee clients.
8:   for  $k \in S_c^t \setminus \{p\}$  do
9:     The  $k$ -th replicate committee client receives the request from  $p$ .
10:    if  $S^k == S^p$  then
11:      The  $k$ -th committee client performs the operation  $o$ .
12:      Return  $\langle \text{Reply}, \text{timestamp}, S^p, \text{response} \rangle$  to  $p$ .
13:    end if
14:    if  $p$  receives at least  $\lfloor C/2 \rfloor + 1$  response then
15:      Consensus is reached on  $S$ .
16:    end if
17:  end for
18: end while

```

# Training Algorithm



- Random Sampling
- Local Training
- Gradient Verification
- Global Model Updating
- Election of New Committee Members

---

**Algorithm 3** The training algorithm of CMFL

---

**Input:**  $\tau, T, K, m, C, \eta_t, \eta_i^t$

**Output:** target global model  $\mathbf{w}^T$

- 1: Initialize  $\mathbf{w}^1, S_c^1$  and  $S_b^1$  randomly.
  - 2: **for**  $t = 1$  to  $T$  **do**
  - 3:   **for**  $k \in S_b^t \cup S_c^t$  **do**
  - 4:     **for**  $i = 1$  to  $\tau$  **do**
  - 5:       The  $k$ -th client runs the SGD over the local dataset by  $\mathbf{w}_{k,i}^t \leftarrow \mathbf{w}_{k,i-1}^t - \eta_i^t g_k(\mathbf{w}_{k,i-1}^t, \mathcal{B}_{k,i-1}^t)$ .
  - 6:     **end for**
  - 7:   **end for**
  - 8:   **for**  $k \in S_b^t$  **do**
  - 9:     **for**  $c \in S_c^t$  **do**
  - 10:       The  $k$ -th training client send its local gradient  $\hat{g}_k^t$  to the  $c$ -th committee client.
  - 11:       The  $c$ -th committee client assign a score  $\mathcal{P}_k^c$  on the  $k$ -th training client/local gradient according to the scoring system.
  - 12:     **end for**
  - 13:   **end for**
  - 14:   The committee clients execute CCP to complete the selection, aggregation, and election process.
  - 15:   The  $S_b^t$  be reinitialized to form the  $S_b^{t+1}$ .
  - 16:   **for**  $k \in S_b^{t+1} \cup S_c^{t+1}$  **do**
  - 17:     The  $k$ -th client downloads the global model from the primary committee client.
  - 18:   **end for**
  - 19: **end for**
-

# Outline



- Background
- Problem Formulation
- Framework
- Theoretical Analysis
- Experimental Results
- Proof of Theorem
- Conclusion

# Assumption



- Assumption 1 and 2 are standard conditions in FL setting

**Assumption 1.** (Lipschitz Gradient).  $F_1, \dots, F_K$  are all  $\mathcal{L}$ -smooth: for all  $\mathbf{v}, \mathbf{w} \in R^n$ ,  $k = 1, \dots, K$ ,  $F_k(\mathbf{v}) \leq F_k(\mathbf{w}) + (\mathbf{v} - \mathbf{w})^T \nabla F_k(\mathbf{w}) + \frac{L}{2} \|\mathbf{v} - \mathbf{w}\|_2^2$ .

**Assumption 2.** ( $\mu$ -strongly Convex Gradient).  $F_1, \dots, F_K$  are all  $\mu$ -strongly convex: for all  $\mathbf{v}, \mathbf{w} \in R^n$ ,  $k = 1, \dots, K$ ,  $F_k(\mathbf{v}) \geq F_k(\mathbf{w}) + (\mathbf{v} - \mathbf{w})^T \nabla F_k(\mathbf{w}) + \frac{\mu}{2} \|\mathbf{v} - \mathbf{w}\|_2^2$

# Assumption



- Assumption 3 is a form of bounded variance between the local objective functions over full dataset and the batch dataset

***Assumption 3.** (Bounded Variance). For the mini-batch  $B_{k,i}^t$  uniformly sampled randomly from  $k$ -th client's dataset  $\mathcal{D}_k$ , the resulting stochastic gradient is unbiased:  $E[g_k(\mathbf{w}_{k,i}^t, \mathcal{B}_{k,i}^t)] = \nabla F_k(\mathbf{w}_{k,i}^t)$  for all  $k = 1, \dots, K, t = 1, \dots, T, i = 1, \dots, \tau$ . And the variance of stochastic grandient in each client is bounded:  $E\|g_k(\mathbf{w}_{k,i}^t, \mathcal{B}_{k,i}^t) - \nabla F_k(\mathbf{w}_{k,i}^t)\|^2 \leq \sigma^2$ .*

# Assumption



- Assumption 4 is fairly standard in nonconvex optimization literature

*Assumption 4. (Bounded Gradient). The local gradient's expected squared norm is uniformly bounded:*  
$$\mathbb{E} \|g_k(\mathbf{w}_{k,i}^t, \mathcal{B}_{k,i}^t)\|^2 \leq G^2 \text{ for all } k = 1, \dots, K, t = 1, \dots, T, i = 1, \dots, \tau.$$

# Assumption



- Assumption 5 is used to constrain the optimal client set and the committee client set caused by the committee mechanism

**Assumption 5.** (Bounded Objective Function). For any aggregation client set  $S_a \notin \emptyset$  and the optimal committee client set  $S_c^* \notin \emptyset$ , the difference of local optimal objective function is bounded:  $\mathbb{E}[|| \sum_{k \in S_a} p_{k, S_a} F_k^* - \sum_{k' \in S_c^*} p_{k', S_c^*} F_{k'}^* ||] \leq \kappa^2$ , where  $S_c^*$  satisfies that  $S_c^* = \arg \min_{S_c} \sum_{k \in S_c} p_{k, S_c} F_k^*$  and  $|S_c^*| = C$ .

# Definition



- The definition of data heterogeneity

**Definition 1.** (*Degree of Heterogeneity*). We use

$$\Gamma = F^* - \sum_{k=1}^K p_k F_k^* = \sum_{k=1}^K p_k (F_k(\mathbf{w}^*) - F_k(\mathbf{w}_k^*))$$

*to quantify the degree of heterogeneity among the clients.*

- In the Non-IID FL scenario, the  $\Gamma$  remains nonzero and its value reflects the heterogeneity of the data distribution. In the IID FL scenario, with the growth of  $K$  it gradually goes to zero

# Definition



- The definition of selection gap between aggregation set and committee set caused by committee mechanism

**Definition 2. (Aggregation-Committee Gap).** For any aggregation client set  $S_a^t$ , we define

$$\begin{aligned} & \varphi(S_a^t, \mathbf{w}) \\ &= \frac{\mathbb{E}[\sum_{k \in S_a^t} p_{k, S_a^t} F_k(\mathbf{w}) - \sum_{k' \in S_c^*} p_{k', S_c^*} F_{k'}^*]}{F(\mathbf{w}) - \sum_{k=1}^K p_k F_k^*} \geq 0, \end{aligned} \quad (10)$$

where  $\mathbb{E}$  denotes the expectation over all randomness in the previous iterations, and  $S_c^*$  denotes the optimal committee client set.

$$\varphi_{min} = \min_{S_a^t, \mathbf{w}} \varphi(S_a^t, \mathbf{w}), \varphi_{max} = \max_{S_a^t} \varphi(S_a^t, \mathbf{w}^*).$$

# Convergence Result



## ■ Analysis

- Selection strategy  $\mathbf{I}$  results in low  $\varphi_{\min}$  and slow convergence rate
- Selection strategy  $\mathbf{\Pi}$  results in high  $\varphi_{\min}$  and fast convergence rate

**Theorem 1.** (Convergence of Committee Mechanism based Federated Learning). Under Assumption 1 to 5, Definition 1 to 2 and the learning rate  $\eta_t$ , where  $\eta_t = \frac{1}{\mu(t+\gamma)}$  and  $\gamma = \frac{4L}{\mu}$ , the error after  $T$  rounds of CMFL satisfies

$$\begin{aligned} & \mathbb{E}[F(\bar{\mathbf{w}}^T)] - F^* \\ & \leq \frac{1}{T + \gamma} \left[ \frac{4L(32\tau^2 G^2 + \sum_{k=1}^K p_k \sigma_k^2) + 24L^2 \kappa^2}{3\mu^2 \varphi_{\min}} \right. \\ & \quad \left. + \frac{8L^2 \Gamma}{\mu^2} + \frac{L\gamma \|\bar{\mathbf{w}}^1 - \mathbf{w}^*\|^2}{2} \right] + \frac{8L\Gamma}{3\mu} \left( \frac{\varphi_{\max}}{\varphi_{\min}} - 1 \right), \end{aligned} \quad (12)$$

where  $L$ ,  $\mu$  and  $\sigma$  represent the constant light indicated in the Assumption 1, 2 and 3.  $G$  and  $\kappa$  represent the upper bound values defined in Assumption 4 and 5.  $\Gamma$  denotes the heterogeneity among the clients according to the Definition

# Computation Complexity



## ■ Local Training

- Computation overhead:  $O((N_t + N_c) |D| \bullet |w|)$
- Computation time:  $O(|D| \bullet |w|)$

## ■ Scoring

- Computation overhead:  $O(N_t \bullet N_c \bullet |w|^2)$
- Computation time:  $O(N_t \bullet |w|^2)$

## ■ Aggregation

- Computation overhead:  $O(m \bullet N_c \bullet |w|)$
- Computation time:  $O(m \bullet |w|)$

## ■ Other computation overhead can be ignored

# Communication Complexity



- Gradient Uploading
  - Communication overhead:  $O(N_t \bullet N_c \bullet |\mathbf{w}|)$
  - Communication time:  $O(N_c \bullet |\mathbf{w}| / r)$
- Global Model Downloading
  - Communication overhead:  $O(N_t \bullet |\mathbf{w}|)$
  - Communication time:  $O(N_t \bullet |\mathbf{w}| / r)$
- Other communication overhead can be ignored

# Outline



- Background
- Problem Formulation
- Framework
- Theoretical Analysis
- Experimental Results
- Proof of Theorem
- Conclusion

# Experimental Settings

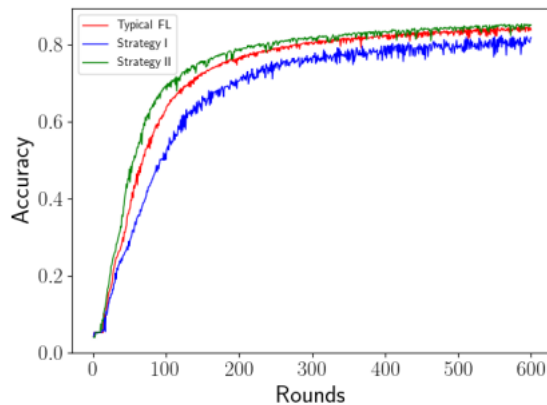


- Dataset & Model
  - FEMNIST-AlexNet
  - Sentiment140-LSTM
- Baseline methods
  - Typical Federated Learning
  - Median
  - Trimmed Mean
  - Krum & Multi-Krum
  - BrainTorrent
  - Gossip Federated Learning

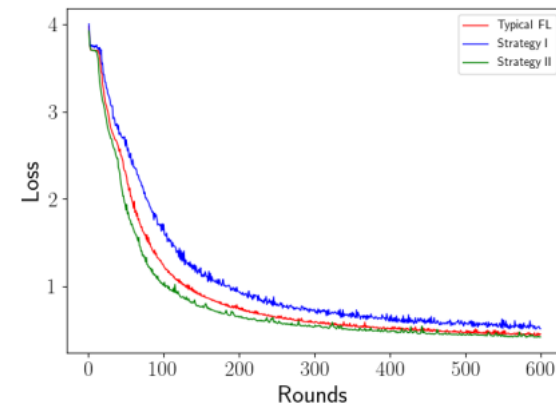
# Normal Training Experiment



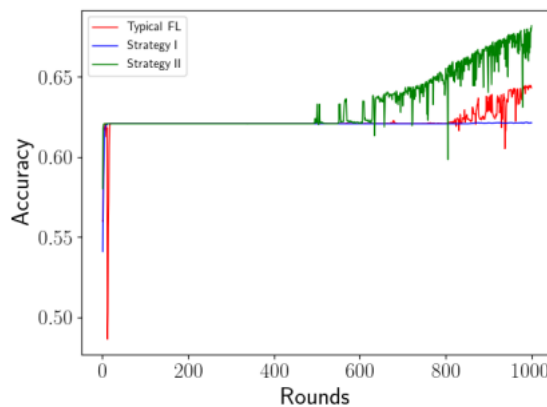
- CMFL under selection strategy  $\Pi$  achieves the best performance in non-attack scenario



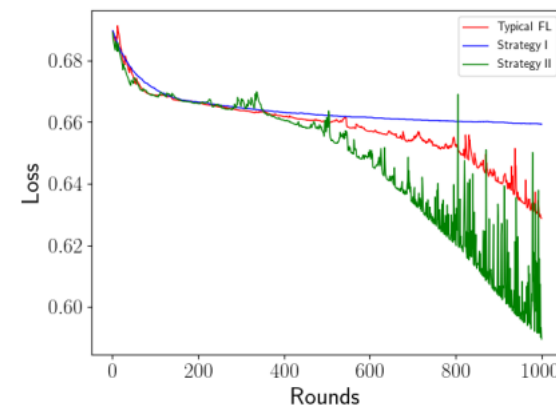
(a) FEMNIST-Accuracy



(b) FEMNIST-Loss



(c) Sentiment140-Accuracy

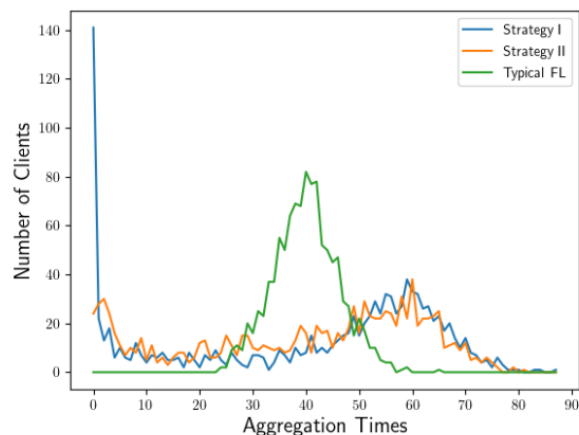


(d) Sentiment140-Loss

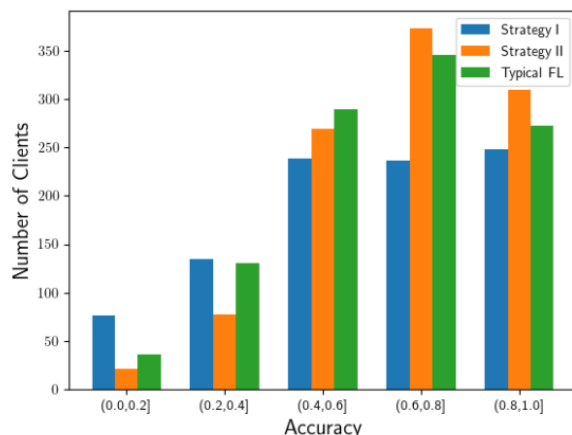
# Normal Training Experiment



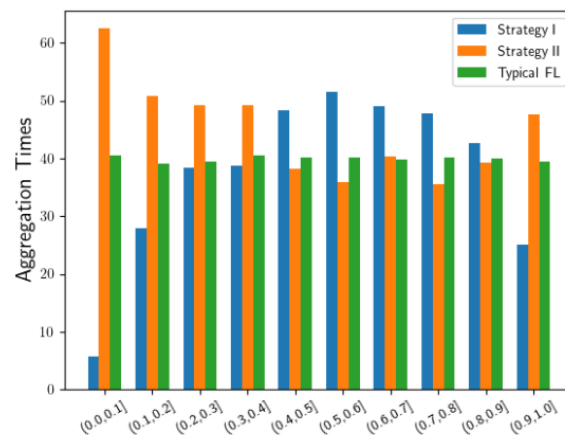
- CMFL under selection strategy  $\Pi$  allows the global model to learn a more comprehensive knowledge



(a)



(b)

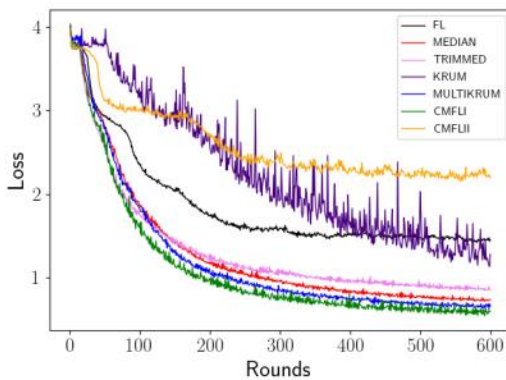


(c)

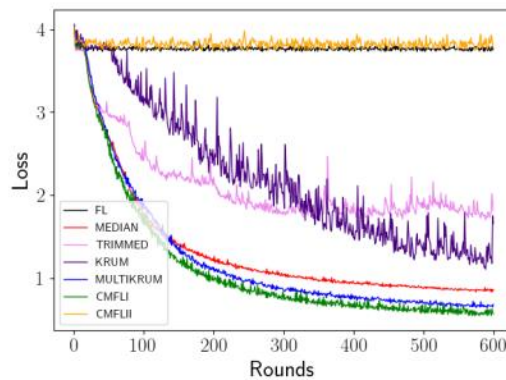
# Robustness Comparative Experiment



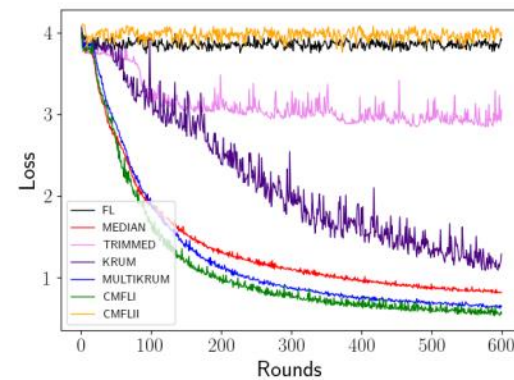
- CMFL under selection strategy I achieves the best performance in Byzantine-attack scenario



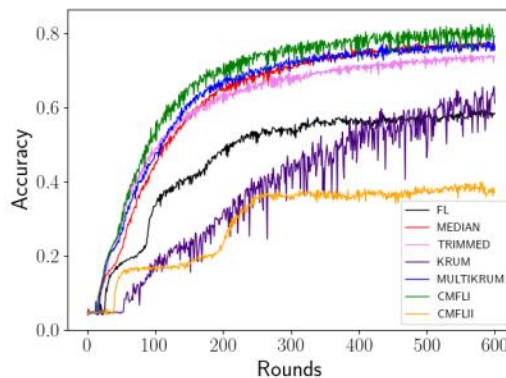
(a) Gradient Scaling Attack Loss



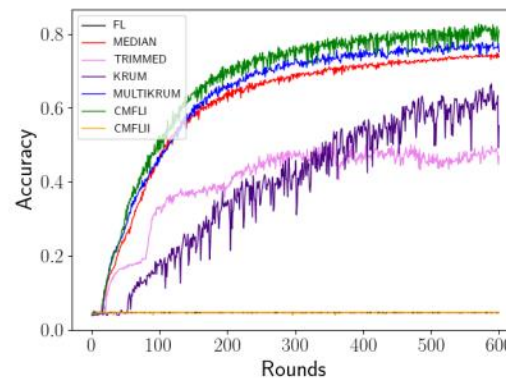
(b) Same-value Attack Loss



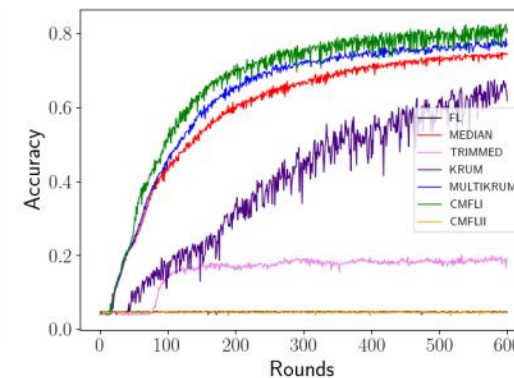
(c) Back-gradient Attack Loss



(d) Gradient Scaling Attack Accuracy



(e) Same-value Attack Accuracy

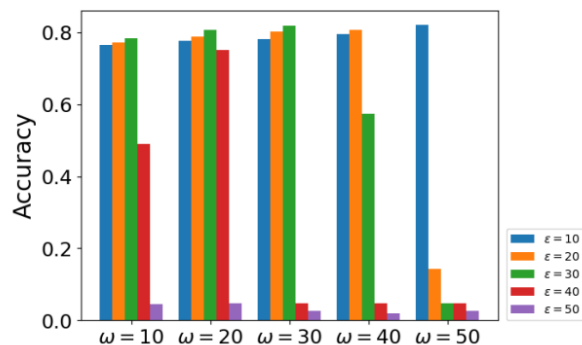


(f) Back-gradient Attack Accuracy

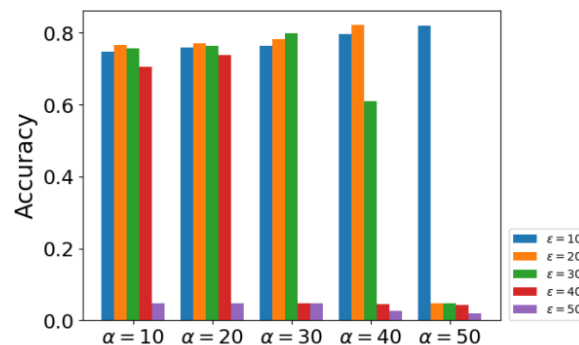
# Hyper-parameter Analysis Experiment



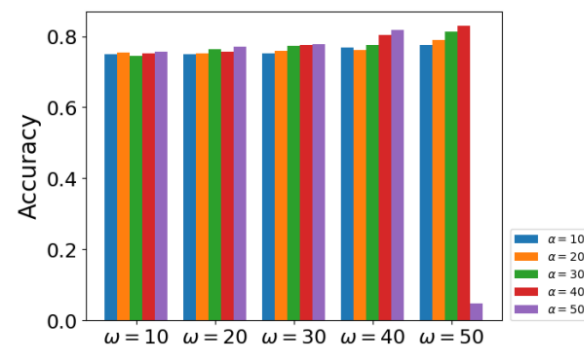
- Extreme parameter settings cause poor performance



(a)  $\alpha = 40$



(b)  $\omega = 40$

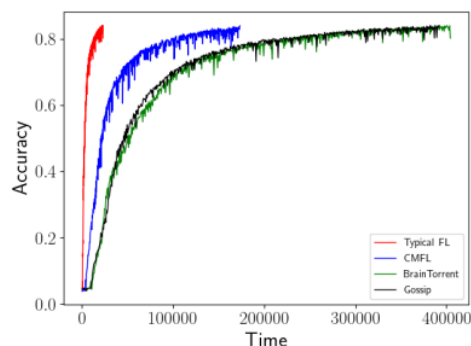


(c)  $\epsilon = 10$

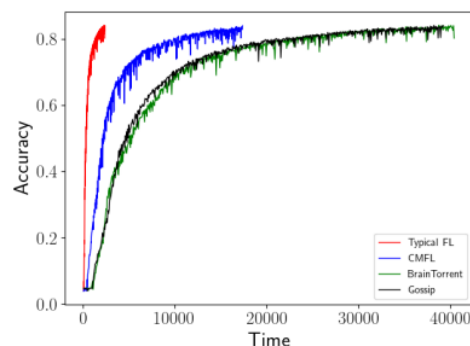
# Efficiency Experiment



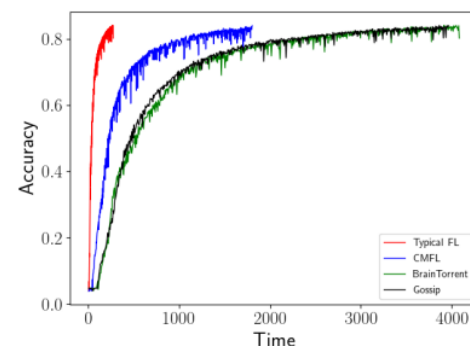
- CMFL achieves a better performance than other two decentralized FL framework (BrainTorrent and Gossip)



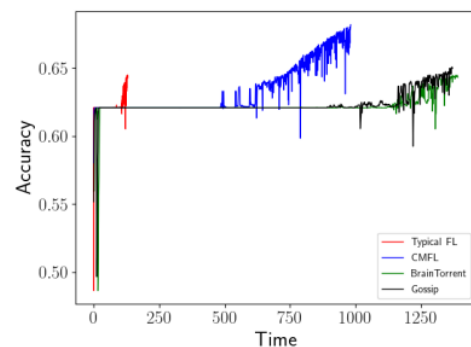
(a) FEMNIST-1Mps



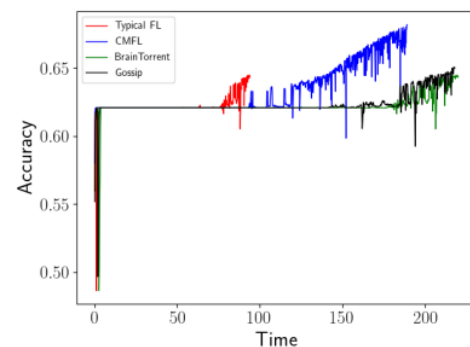
(b) FEMNIST-10Mps



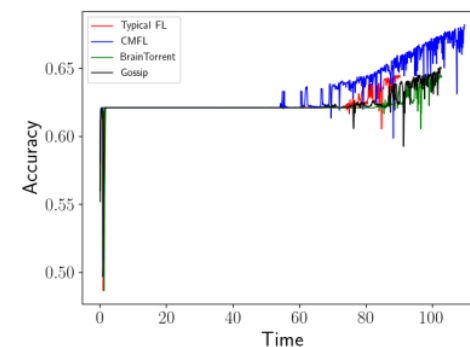
(c) FEMNIST-100Mps



(d) Sentiment140-1Mps



(e) Sentiment140-10Mps

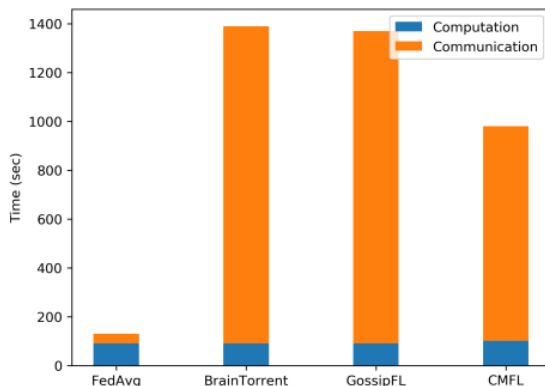


(f) Sentiment140-100Mps

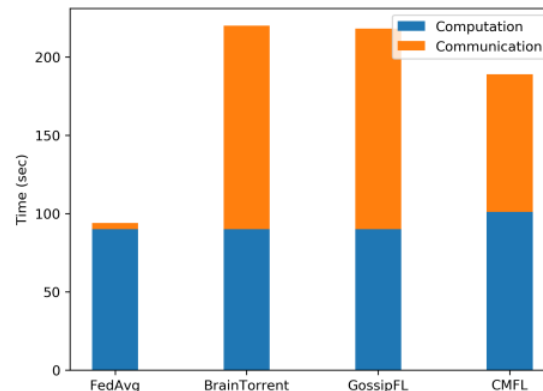
# Efficiency Experiment



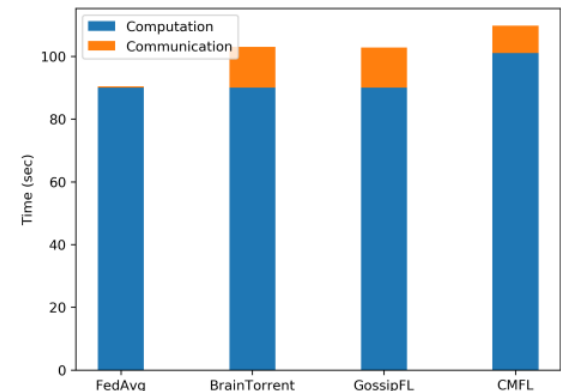
- The communication overhead is lower than other two decentralized FL framework



(a) Sentiment140-1Mps



(b) Sentiment140-10Mps

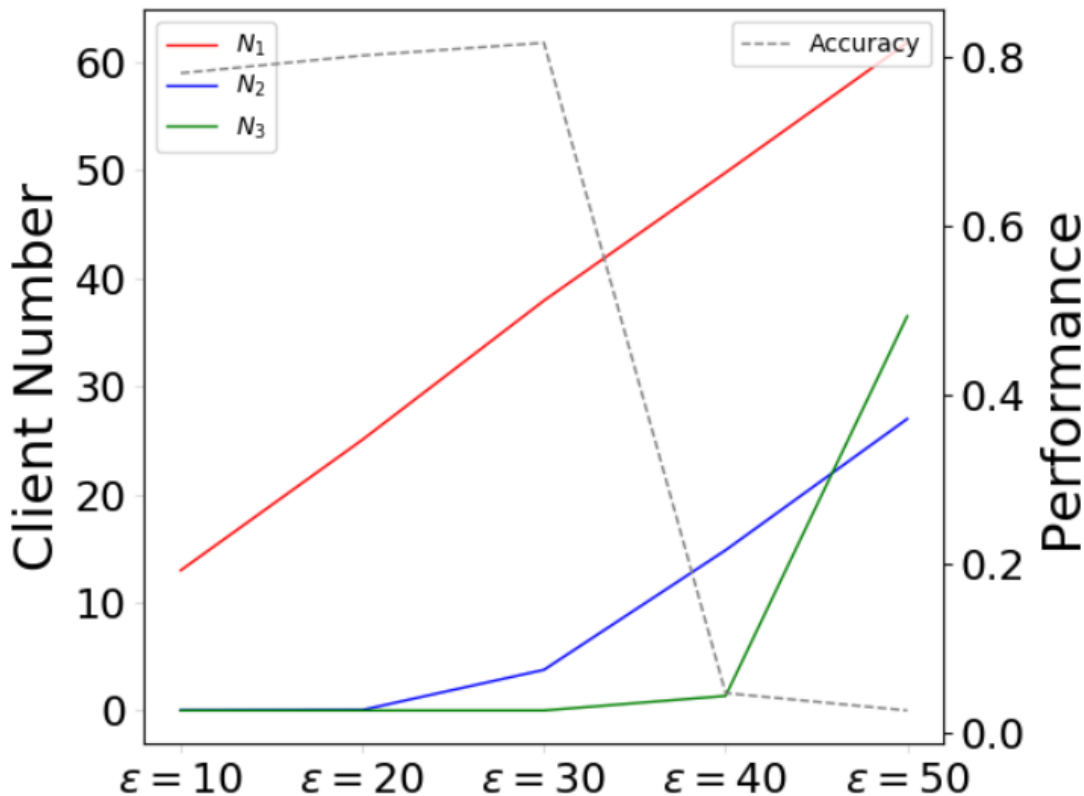


(c) Sentiment140-100Mps

# Committee Member Analysis Experiment



- The committee system is robust enough



# Outline



- Background
- Problem Formulation
- Framework
- Theoretical Analysis
- Experimental Results
- Proof of Theorem
- Conclusion

# Definition



- (t) is defined as the index of the total local SGD iterations
- Local SGD training

$$\mathbf{w}_k^{(t+1)} = \mathbf{w}_k^{(t)} - \eta_t g_k(\mathbf{w}_k^{(t)}, \mathcal{B}_k^{(t)}),$$

- Aggregation

$$\bar{\mathbf{w}}^{(t+1)} = \bar{\mathbf{w}}^{(t)} - \eta_t \bar{g}^{(t)} = \bar{\mathbf{w}}^{(t)} - \eta_t \left( \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} g_k(\mathbf{w}_k^{(t)}, \mathcal{B}_k^{(t)}) \right),$$

- Local Updating

$$\mathbf{w}_k^{(t+1)} = \begin{cases} \mathbf{w}_k^{(t)} - \eta_t g_k(\mathbf{w}_k^{(t)}, \mathcal{B}_k^{(t)}), & (t+1) \% \tau \neq 0 \\ \bar{\mathbf{w}}^{(t+1)}, & (t+1) \% \tau = 0. \end{cases}$$

# Lemma



**Lemma 1. (Global-Local Gradient Product).** According to the definitions of  $\nabla F_k(\mathbf{w}_k^{(t)})$ , we have that

$$\begin{aligned}
 & - \langle \bar{\mathbf{w}}^{(t)} - \mathbf{w}^*, \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)}) \rangle \\
 & \leq \frac{1}{2\eta_t} \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}_k^{(t)}\|^2 + \frac{\eta_t}{2} \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\nabla F_k(\mathbf{w}_k^{(t)})\|^2 - \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}_k^{(t)}) - F_k(\mathbf{w}^*)) \\
 & \quad - \frac{\mu}{2} \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\mathbf{w}_k^{(t)} - \mathbf{w}^*\|^2.
 \end{aligned}$$

**Lemma 2. (Local Parameter-Optimal Gap).** Let  $v_t = 2\eta_t - 4L\eta_t^2 = 2\eta_t(1 - 2L\eta_t)$ , we have that

$$\begin{aligned}
 & - \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}_k^{(t)}) - F_k^*) \\
 & \leq - (1 - \eta_t L) \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\bar{\mathbf{w}}^{(t)}) - F^*) + \frac{1}{v_t} \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\mathbf{w}_k^{(t)} - \bar{\mathbf{w}}^{(t)}\|^2.
 \end{aligned}$$

**Lemma 3. (Client Heterogeneous Bound).** According to the Assumption 4, we have that

$$\mathbb{E} \left[ \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}_k^{(t)}\|^2 \right] \leq 16\eta_t^2 \tau^2 G^2.$$

# Proof of Theorem 1



- According to the global updating:

$$\mathcal{H}(\mathbf{w}, t+1) = \|\bar{\mathbf{w}}^{(t+1)} - \mathbf{w}^*\|^2 = \|\bar{\mathbf{w}}^{(t)} - \eta_t \bar{g}^{(t)} - \mathbf{w}^*\|^2.$$

- Above formula can be expanded as:

$$\begin{aligned} & \mathcal{H}(\mathbf{w}, t+1) \\ = & \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^* - \eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)})\|^2 + \|\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)}) - \eta_t \bar{g}^{(t)}\|^2 \\ & + 2\eta_t \underbrace{\langle \bar{\mathbf{w}}^{(t)} - \mathbf{w}^* - \eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)}), \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)}) - \bar{g}^{(t)} \rangle}_{A_1}. \end{aligned}$$

# Proof of Theorem 1



- Due to Assumption 3 and Lemma 1:

$$\mathcal{H}(\mathbf{w}, t+1)$$

$$\begin{aligned} &\leq \underbrace{\|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^*\|^2 + 4L\eta_t^2 \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}_k^{(t)}) - F_k^*)}_{A_2} - 2\eta_t \underbrace{\sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}_k^{(t)}) - F_k(\mathbf{w}^*))}_{A_3} \\ &\quad + \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}_k^{(t)}\|^2 - \eta_t \mu \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\mathbf{w}_k^{(t)} - \mathbf{w}^*\|^2 + \|\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)}) - \eta_t \bar{g}^{(t)}\|^2 + A_1. \end{aligned}$$

- The difference of A2 and A3 can be written as:

$$\begin{aligned} &A_2 - A_3 \\ &= (4L\eta_t^2 - 2\eta_t) \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k(\mathbf{w}_k^{(t)}) - 4L\eta_t^2 \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k^* + 2\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k(\mathbf{w}^*) \\ &= (4L\eta_t^2 - 2\eta_t) \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k(\mathbf{w}_k^{(t)}) - (4L\eta_t^2 - 2\eta_t) \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k^* + 2\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k(\mathbf{w}^*) - 2\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k^* \\ &= (4L\eta_t^2 - 2\eta_t) \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}_k^{(t)}) - F_k^*) + 2\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}^*) - F_k^*). \end{aligned}$$

# Proof of Theorem 1



- Let  $v_t = 2\eta_t - 4L\eta_t^2 = 2\eta_t(1 - 2L\eta_t)$  :

$$\begin{aligned}
 & \mathcal{H}(\mathbf{w}, t+1) \\
 & \leq \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^*\|^2 - v_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}_k^{(t)}) - F_k^*) + 2\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}^*) - F_k^*) + \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}_k^{(t)}\|^2 \\
 & \quad - \eta_t \mu \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\mathbf{w}_k^{(t)} - \mathbf{w}^*\|^2 + \|\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)}) - \eta_t \bar{g}^{(t)}\|^2 + A_1.
 \end{aligned}$$

- Due to Lemma 2:

$$\begin{aligned}
 & \mathcal{H}(\mathbf{w}, t+1) \\
 & \leq \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^*\|^2 + 2\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}^*) - F_k^*) + 2 \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}_k^{(t)}\|^2 \\
 & \quad - v_t(1 - \eta_t L) \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\bar{\mathbf{w}}^{(t)}) - F_k^*) - \eta_t \mu \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\mathbf{w}_k^{(t)} - \mathbf{w}^*\|^2 \\
 & \quad + \|\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)}) - \eta_t \bar{g}^{(t)}\|^2 + A_1.
 \end{aligned}$$

# Proof of Theorem 1



## ■ Solve the expectation:

$$\begin{aligned}
 \mathbb{E}[\mathcal{H}(\mathbf{w}, t+1)] &= \mathbb{E}[\|\bar{\mathbf{w}}^{(t+1)} - \mathbf{w}^*\|^2] \\
 &\leq \mathbb{E}[\|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^*\|^2] + \mathbb{E}[2\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}^*) - F_k^*)] + \mathbb{E}[2 \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}_k^{(t)}\|^2] \\
 &\quad - \mathbb{E}[v_t(1 - \eta_t L) \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\bar{\mathbf{w}}^{(t)}) - F_k^*)] - \mathbb{E}[\eta_t \mu \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\mathbf{w}_k^{(t)} - \mathbf{w}^*\|^2] \\
 &\quad + \mathbb{E}[\|\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \nabla F_k(\mathbf{w}_k^{(t)}) - \eta_t \bar{g}^{(t)}\|^2] + \mathbb{E}[A_1].
 \end{aligned}$$

## ■ Due to Assumption 3 and the expectation of A1 is zero:

$$\begin{aligned}
 &\mathbb{E}[\|\bar{\mathbf{w}}^{(t+1)} - \mathbf{w}^*\|^2] \\
 &= (1 - \eta_t \mu) \mathbb{E}[\|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^*\|^2] + \mathbb{E}[\mathcal{Q}(\mathbf{w}, k, t)] + \eta_t^2 \sum_{k=1}^K p_k^2 \sigma_k^2,
 \end{aligned}$$

where

$$\begin{aligned}
 &\mathcal{Q}(\mathbf{w}, k, t) \\
 &= 2\eta_t \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\mathbf{w}^*) - F_k^*) + 2 \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} \|\bar{\mathbf{w}}^{(t)} - \mathbf{w}_k^{(t)}\|^2 - v_t(1 - \eta_t L) \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} (F_k(\bar{\mathbf{w}}^{(t)}) - F_k^*).
 \end{aligned}$$

# Proof of Theorem 1



- Due to Lemma 3:

$$\begin{aligned}
 & \mathbb{E}[\mathcal{Q}(\mathbf{w}, k, t)] \\
 &= -v_t(1 - \eta_t L) \mathbb{E} \left[ \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k(\bar{\mathbf{w}}^{(t)}) - \sum_{k' \in S_c^*} p_{k', S_c^*} F_{k'}^* \right] + 2\eta_t \left( \mathbb{E} \left[ \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k(\mathbf{w}^*) - \sum_{k' \in S_c^*} p_{k', S_c^*} F_{k'}^* \right] \right. \\
 & \quad \left. - (2\eta_t - v_t(1 - \eta_t L)) \mathbb{E} \left[ \sum_{k \in S_a^{(t)}} p_{k, S_a^{(t)}} F_k^* - \sum_{k' \in S_c^*} p_{k', S_c^*} F_{k'}^* \right] + 32\eta_t^2 \tau^2 G^2 \right).
 \end{aligned}$$

- According to the Assumption 5 and Definition 1 and 2:

$$\begin{aligned}
 & \mathbb{E}[\mathcal{Q}(\mathbf{w}, k, t)] \\
 & \leq \underbrace{-v_t(1 - \eta_t L) \varphi_{\min} \mathbb{E} \left[ (F(\bar{\mathbf{w}}) - \sum_{k=1}^K p_k F_k^*) \right]}_{A_4} + 2\eta_t \varphi_{\max} \Gamma + 6L\eta_t^2 \kappa^2 + 32\eta_t^2 \tau^2 G^2.
 \end{aligned}$$

- A4 can be expanded as:

$$A_4 \leq -\frac{3\eta_t \mu \varphi_{\min}}{8} \mathbb{E}[\|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^*\|^2] - 2\eta_t \varphi_{\min} \Gamma + 6\eta_t^2 \varphi_{\min} L \Gamma.$$

# Proof of Theorem 1



So we have

$$\begin{aligned} & \mathbb{E}[Q(\mathbf{w}, k, t)] \\ &= -\frac{3\eta_t\mu\varphi_{\min}}{8}\mathbb{E}[\|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^*\|^2] + 2\eta_t\Gamma(\varphi_{\max} - \varphi_{\min}) + \eta_t^2(6\varphi_{\min}L\Gamma + 32\tau^2G^2 + 6L\kappa^2). \end{aligned} \quad (34)$$

As a result, we have

$$\begin{aligned} & \mathbb{E}[\|\bar{\mathbf{w}}^{(t+1)} - \mathbf{w}^*\|] \\ & \leq [1 - \eta_t\mu(1 + \frac{3\varphi_{\min}}{8})]\mathbb{E}[\|\bar{\mathbf{w}}^{(t)} - \mathbf{w}^*\|^2] + 2\eta_t\Gamma(\varphi_{\max} - \varphi_{\min}) + \eta_t^2(6\varphi_{\min}L\Gamma + 32\tau^2G^2 + 6L\kappa^2 + \sum_{k=1}^K p_k\sigma_k^2). \end{aligned} \quad (35)$$

By defining  $\Delta_{t+1} = \mathbb{E}[\|\bar{\mathbf{w}}^{(t+1)} - \mathbf{w}^*\|]$ ,  $B = 1 + \frac{3\varphi_{\min}}{8}$ ,  $C = 6\varphi_{\min}L\Gamma + 32\tau^2G^2 + 6L\kappa^2 + \sum_{k=1}^K p_k\sigma_k^2$ ,  $D = \Gamma(\varphi_{\max} - \varphi_{\min})$ , we have

$$\Delta_{t+1} \leq (1 - \eta_t\mu B)\Delta_t + \eta_t^2 C + \eta_t D. \quad (36)$$

If we set  $\Delta_t \leq \frac{\psi}{t+\gamma}$ ,  $\eta_t = \frac{\beta}{t+\gamma}$  and  $\beta > \frac{1}{\mu B}$ ,  $\gamma > 0$  by induction, we have

$$\psi = \max \left\{ \gamma \|\bar{\mathbf{w}}^1 - \mathbf{w}^*\|^2, \frac{1}{\beta\mu B - 1}(\beta^2 C + D\beta(t + \gamma)) \right\}. \quad (37)$$

Then by the L-smoothness of  $F(\cdot)$ ,

$$\mathbb{E}[F(\bar{\mathbf{w}}^{(t)})] - F^* \leq \frac{L}{2}\Delta_t \leq \frac{L}{2} \frac{\psi}{\gamma + t}. \quad (38)$$

Finally, we complete the proof of Theorem 1:

$$\begin{aligned} & \mathbb{E}[F(\bar{\mathbf{w}}^T)] - F^* \\ & \leq \frac{1}{T + \gamma} \left[ \frac{4L(32\tau^2G^2 + \sum_{k=1}^K p_k\sigma_k^2) + 24L^2\kappa^2}{3\mu^2\varphi_{\min}} + \frac{8L^2\Gamma}{\mu^2} + \frac{L\gamma\|\bar{\mathbf{w}}^1 - \mathbf{w}^*\|^2}{2} \right] + \frac{8L\Gamma}{3\mu} \left( \frac{\varphi_{\max}}{\varphi_{\min}} - 1 \right), \end{aligned} \quad (39)$$

where the  $T$  means the maximal communication rounds, which satisfies  $T = i\tau$  for  $i = 1, 2, \dots$  in realistic scenarios.  $\square$

# Outline



- Background
- Problem Formulation
- Framework
- Theoretical Analysis
- Experimental Results
- Proof of Theorem
- Conclusion

# Conclusion



## ■ Advantages

- Strong Robustness
- Decentralized Setting
- High Communication Efficiency
- Theoretical Guarantee
- Scalable Framework
- Better Performance on Non-IID dataset

## ■ Future Works

- Backdoor attack
- System Heterogeneity

**Thank You!**  
**Q & A**